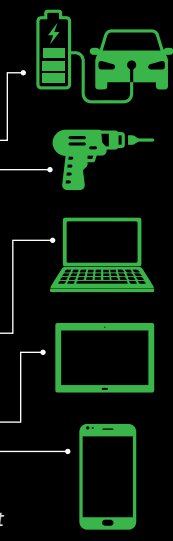
LITHIUM IN BATTERIES

How much lithium is in everyday usage



Source: Visualcapitalist
**Approx. weight*

Rs 18,100 crore incentive to boost the manufacturing of batteries for EVs

Last year, India signed a pact with an Argentinian firm to jointly prospect lithium in the South American nation, which has 17 MT of reserves, through the state-owned Khanij Bidesh India Ltd

GSI report. Amit Sharma, secretary, mining department, says they will take further decisions about the exploration process in consultation with the central government. "The auction of mines will come at a later stage," he maintains.

Another major factor that contributes to the challenges is the technology needed for lithium exploration. The Salal find suggests a deposit of bauxite ore rich in lithium and will need separation technology to extract lithium from the ore. Such a technology, according to Prof. Srivastava, is used in Chile or Argentina where lithium is found in clay minerals, but here it needs to be seen how much lithium can be recovered and whether it is worth the cost. "Beneficiation studies, which are yet to be conducted, can give grade and tenor of the mineral. Also, it can give quantity and quality as well," the geologist adds. "It needs an electric method to separate lithium from bauxite and hence power supply is critical. But it takes a long time to reach that G1 stage wherein the ore body has to be opened and people have to be relocated as happens in any other mineral block."

Spread over 1,719 square kilometres, the hilly Reasi district falling amid ravines and gorges has 67,498 hectares of land under forest cover. Some fear that mining would play havoc with the ecology and might even trigger quakes in the region, which lies in high-risk seismic zones. In October 2005, an earthquake of 7.6 magnitudes on the Richter scale caused massive destruction in J&K. In 2016, an Oregon State University (US) study revealed that the region was ripe for a major earthquake that could endanger the lives of as many as a million people. They did a fresh geologic mapping of the Himalayan mountains, which showed that the Reasi fault had been building up pressure for some time, suggesting that when it does release or 'slip', the resulting earthquake may hit 8 or higher on the Richter scale. Yann Gavillot, who led the study as a doctoral student, said, "What we found was that the Reasi fault is one of the main active faults in Kashmir but there is a lack of earthquakes in the more recent geologic record. The fault hasn't slipped for a long time, which means the potential for a large earthquake is strong. It's not a question of if it's going to happen. It's a matter of when."

FOR NOW, THE J&K GOVERNMENT HAS ASSIGNED A TEAM OF EXPERTS TO STUDY THE GSI REPORT

Scientists like Prof. Srivastava agree that mining results in environmental degradation everywhere, including the Himalayas, and it results in the displacement of people. However, he dismisses the suggestions that mining can trigger quakes.

The discovery has sent a rush of excitement in the 24.68 square kilometre territory of Salal village and its 600-700 families. Their only concern is that they should be adequately compensated and resettled together in a place where they can continue to share the centuries-old sense of community that binds this Hindu-majority village. "We are happy as well as sad. This is not a village; we are one big family. In happiness and grief, we are together. But the mining will uproot us and leave us disintegrated," says Mahatam Singh, a resident of the village.

Until the 1980s when the Salal power project was set up, farming was the mainstay of their livelihood. Now the villagers expect the hill on which their houses and farms stand to transform their fortunes. It is not the first time they have heard about valuable minerals on their land. Almost four decades ago, a firm from Odisha came to the village to extract rock samples. They told the villagers they contained gold, silver and bauxite. But they heard no more of it until the GSI turned up for a reconnaissance survey in 1996.

Deputy sarpanch Rajinder Singh only wishes that the government had pursued the operation years ago, saving people from relocation in their old age. "At this stage," he says, "when most of my contemporaries are in their 50s, displacement will be cruel. We have no option but to accept the government's order." However, these fears—and hopes—are based on speculation as things stand. There has been no word from the administration about any plans to relocate the community.

Nonetheless, the villagers plan a grand meeting to decide a strategy for the future. Their excitement, or anxiety, will keep brewing till the time the lithium find is confirmed. ■

with Amarnath K. Menon